

Maintenance Wizard Report

Executive Summary

CCM Segment Bearing is exhibiting critical symptoms of advanced failure, including high vibration, increasing temperature, and abnormal noise. This indicates a high probability of imminent catastrophic failure with significant production and safety risks.

Diagnosis

Advanced stage bearing failure within the CCM segment. Symptoms strongly suggest severe damage to bearing races, rolling elements, or cage, likely due to fatigue, lubrication breakdown, or contamination ingress.

Root Cause

Preliminary assessment suggests the root cause is likely a combination of inadequate lubrication, contamination (e.g., water, scale, or metallic particles), excessive loading, or long-term material fatigue. Misalignment or improper installation during previous maintenance cannot be ruled out without a detailed post-mortem analysis.

Risk Level

CRITICAL - High. Imminent risk of catastrophic failure, leading to potential secondary damage to the CCM segment structure, roll deflection, breakout, significant production downtime, and safety hazards.

Failure Probability

Imminent - Very High. Given the rapid progression of symptoms, complete failure is expected within a very short operational period.

Remaining Useful Life

Less than 24-48 hours under current operating conditions. Continued operation significantly increases risk of catastrophic failure.

Catastrophic Risk

Bearing seizure, resulting in damage to the segment housing, rolls, and potential for a hot metal breakout. Extensive secondary damage will lead to prolonged unplanned downtime, high repair costs, and significant production losses. High safety risk to personnel due to flying debris or hot metal.

Immediate Actions

[Immediately inform Production and Operations Management about the critical status and potential need for an emergency shutdown.], 'Initiate immediate preparations for a controlled shutdown of the

casting machine.', 'Monitor bearing temperature and vibration continuously, adjusting casting speed and load to minimize stress if shutdown is not immediate.', 'Assemble maintenance crew and prepare necessary tools and safety equipment for segment replacement.', 'Verify availability of the specific CCM Segment Bearing spare part (P/N: XXXXXX) and associated seals/fasteners.']

Short Term Actions

['Execute planned emergency shutdown and replace the faulty CCM segment bearing immediately.', 'Perform visual inspection of the adjacent segments and rolls for any signs of premature wear or damage during the shutdown.', 'Conduct a detailed post-mortem analysis of the failed bearing to ascertain the precise root cause (metallurgical analysis, lubrication inspection, wear pattern analysis).', 'Review the lubrication system for the CCM segments, including grease type, frequency, and delivery pressure/volume.']

Long Term Actions

['Implement or enhance online condition monitoring (vibration analysis, thermography) for all critical CCM segment bearings.', 'Review and update standard operating procedures (SOPs) for bearing installation and lubrication schedules.', 'Investigate potential sources of contamination ingress (seal effectiveness, water spray systems).', 'Evaluate alternative bearing designs or lubrication systems for improved robustness in the harsh steel plant environment.', 'Establish a robust spare parts inventory and procurement strategy for critical components like CCM segment bearings.']

Spare Procurement

['CCM Segment Bearing (Exact P/N: [To be confirmed based on segment drawing])', 'Associated seals and gaskets (P/N: [To be confirmed])', 'Mounting hardware (bolts, nuts, washers - specific grades)', 'Specialized high-temperature, water-resistant grease (e.g., [Brand/Type])']

Maintenance Recommendation

Urgent and immediate replacement of the failing CCM segment bearing is required to prevent catastrophic failure, minimize secondary damage, and ensure operational safety. A comprehensive root cause analysis must follow the replacement to implement effective preventive measures and improve overall CCM reliability.

Conclusion

The CCM segment bearing is in a critical state of failure, demanding immediate shutdown and replacement. Failure to act swiftly will lead to severe operational and safety consequences. This incident highlights the need for a more robust predictive maintenance strategy and thorough root cause investigations to prevent recurrence.